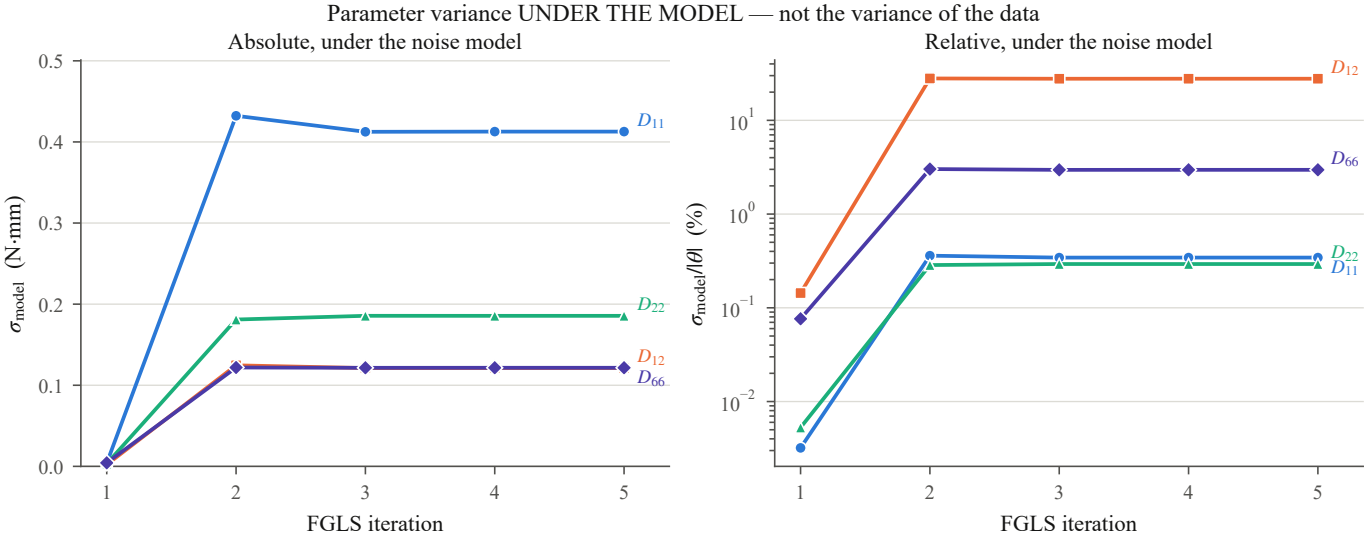

These curves are eq. (20) evaluated on eq. (3) exactly as derived ($\kappa = 1$; the measured spatial correlation is reported in fig. D but never folded in).

They answer: "if the ONLY error were spatially white DIC noise with covariance $\hat{\Gamma}$, how much would $\hat{\theta}$ move?" They do NOT answer "how well is D_{ij} known?".

The data reject that noise model at $\chi^2/\text{dof} = 285$: it accounts for 0.35 % of the observed residual variance. An interval to quote is at least $\sqrt{\chi^2/\text{dof}} = 17\times$ these values, and even that assumes the misfit is a pure scale error rather than a shape error.